MIL-P-63141B (AR)
30 August 1985
SUPERSEDING
MIL-P-63141A (AR)
13 February 1981

MILITARY SPECIFICATION

PROJECTILE, 8-INCH, HE, RA, M650 BOOSTER AUXILIARY PA-500 AND BOOSTER ASSEMBLY PA-524 FOR

This specification is approved for use by US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication and packing of the PA-500 Booster Auxiliary and Booster Assembly PA-524.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATONS

MILITARY

MIL-A-48078 -	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 -	Marking for Shipment and Storage
MIL-STD-453 -	Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1320

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(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.7)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

8861645 - Box, Fiber, Packing, Ammunition, for Supplementary Charge for Deep Cavity Loaded Projectiles or Booster Assembly PA 524
8861646 - Box, Packing, Ammunition, for Supplementary Charge for Deep Cavity Loaded Projectiles or Booster Assembly, PA-524
9287977 - Booster Auxiliary PA-500
9326028 - Packing and Marking For Box, Wood For Booster, Auxiliary PA-500
9370179 - Booster Assembly PA-524
9370180 - Pellet

Code of Federal Regulations

Title 49 - Transportation parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)").

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Booster auxiliary PA-500. The booster assembly shall comply with all requirements specified on Drawing 9287977, component drawings and all applicable specifications.

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3.3 Radiographic examination of PA-500 booster assembly. The loaded booster assembly shall be radiographically examined for proper assembly and presence of foreign material, including proper seating of the liner and presence of only one liner.

3.4 Functioning test of PA-500. The booster assembly shall penetrate 1.0 inches of aluminum when tested in accordance with paragraph 4.4.2.3.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Booster assembly, PA-524. The booster assembly shall comply with all requirements specified on Drawing 9370179, component drawings and all applicable specifications.

3.7 Workmanship. All processing and assembling operations shall be correctly performed in a thorough workmanlike manner. Parts shall be free of burrs, chips, cracks, dirt, grease, rust, corrosion products or other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent. Exterior surface coating shall be continuous except for a few light scratches exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 20 Booster Auxiliary PA-500 Assemblies per Drawing 9287977, 20 Booster Assemblies PA-524 per Drawing 9370179, and 10 Pellets per Drawing 9370180.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63141B (AR)

PARAGRAPH	TITLE		SHEET 1 of 2		DRAWING NUMBER See Below
	Boosters Auxiliary PA-500 and PA-524				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Booster Prior to Assembly of Disk, Closure (Dwg. 9287977) Examination for Defects	20	-	3.2	4.4.2.1
	Booster Prior to Assembly of Washer, Closure (Dwg. 9287977) Examination for Defects	20	-	3.2	4.4.2.2
	Booster Auxiliary PA-500 (Dwg. 9287977) Examination for Defects	20	-	3.2	4.4.2.3
	Pellet (Dwg. 9370180) Examination for Defects	10	-	3.6	4.4.2.4
	Booster Assembly, Prior to Assembly of Closing Disc (Dwg. 9370179) Examination for Defects	20	-	3.6	4.4.2.5

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See Below
	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Booster Assembly, Prior to Assembly of Pad (Dwg. 9370179) Examination for Defects	20	-	3.6	4.4.2.6
	Booster Assembly PA-524 (Dwg. 9370179) Examination for Defects	20	-	3.6	4.4.2.7

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Booster Assemblies shall contain:

a. Inert parts from one interfix lot number from one manufacturer.
b. Explosive composition from not more than one interfix lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63141B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	PA-500 Booster Prior to Assembly of Disk, Closure	1 OF 1	9287977
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Comp A5 missing or improper		0.40%	3.2	Visual
102	Specific gravity of explosive composition A5	a	-	3.2	4.4.3.2/4.5.2/Test
Minor					
201	Evidence of poor workmanship		1.0	3.7	Visual

NOTES: a. See Paragraph 4.4.3.2.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Booster Auxiliary PA-500 Prior to Assembly of Washer, Closure	1 OF 1	9287977
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
Major 101	Disk, Closure mising or improperly assembled		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63141B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Booster Auxiliary PA-500	1 OF 1	9287977
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Washer closure not flush to below flush		0.40%	3.2	Visual/Manual
102	Washer cover not securely staked		0.40%	3.2	Visual/Manual
103	Cover coat missing or improperly applied to washer, closure		0.40%	3.2	Visual
104	Liner inverted or improperly positioned		0.40%	3.2	Visual
105	Improper loading or assembly of booster		100%	3.3	4.4.3.1/4.5.1/Test
106	Functioning test	a	-	3.4	4.4.3.3/4.5.3
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

a. See paragraph 4.4.3.3.

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QUALITY CONFORMANCE INSPCTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63141B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Pellet for PA-524	1 OF 1	9370180
			NEXT HIGHER ASSEMBLY 9370179

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.6	Gage
102	Height		0.40%	3.6	Gage
103	Weight		0.40%	3.6	Balance
104	Density		Note 1	3.6	Gage/Balance
Minor	None defined				

NOTES: 1. Per dwg. 9370180, check density if sample fails to meet weight requirement, but meets requirements for diameter and height.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63141B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	PA-524 Booster Assembly, Prior to Assembly of Closing Disc				9370179
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Any component missing or damaged		0.40%	3.6	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63141B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	PA-524 Booster Assembly, Prior to Assembly of Pad	1 OF 1	9370179
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly height		0.40%	3.6	Gage
102	Crack, split, puncture or similar damage extending through cup or closing disc		100%	3.6	Visual
103	Crimp improper		100%	3.6	Visual
104	Pellets loose		100%	3.6	Manual
Minor					
201	Diameter, max.		0.65%	3.6	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Booster Assembly PA-524	1 OF 1	9370179
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pad missing or insecure		100%	3.6	Visual/Manual
Major	None defined				
Minor					
201	Pad improperly aligned		0.65%	3.6	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing

4.4.3.1 Radiographic examination for improper assembly and foreign material in explosive composition. This test shall be performed on all Booster Auxiliary PA-500 assemblies. The radiographic test shall be performed using equipment in accordance with 4.4.4.

4.4.3.2 Specific gravity of explosive composition. A sample of 5 charges shall be randomly selected from each press from each lot and the specific gravity determined in accordance with 4.5.2 for compliance with the requirement on Drawing 9287977. If any charge does not meet the required specific gravity the lot shall be rejected.

4.4.3.3 Functioning test. A sample of 5 assemblies shall be randomly selected from each press from each lot and tested in accordance with 4.5.3 for compliance with the requirement in 3.4. If any charge fails to meet the requirements the lot shall be rejected.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and Paragraph 6.3 herein.

4.5 Methods of inspection.

4.5.1 Determination of improper assembly or foreign material. The Booster Auxiliary PA-500 shall be subjected to radiographic examination (see 4.4.3.1). In all exposures the X-ray beam shall form a 90° angle with the longitudinal axis of the Booster Case. Examination shall be made to determine compliance with the applicable requirements.

4.5.1.1 Radiographic equipment, operations and procedures. Radiographic equipment, operations and procedures shall be qualified in accordance with standard MIL-STD-453 (see 6.6). In addition, the sensitivity of the x-ray technique shall be 2.0 percent minimum.

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4.5.2 Determination of specific gravity. The specific gravity of the explosive composition shall be calculated as follows:

$$S.G. = \frac{W2 - W1}{(.5401 - .7889L)\ (.0361)}$$

W_1 = weight of case - adapter booster with liner
W_2 = weight of case - adapter booster, liner and explosive composition
L = distance (inch) from open end of case to face of explosive composition

NOTE: If weight is given in grams, find weight in pounds as follows:

$$W = \frac{\text{grams}}{453.6} = \text{lbs.}$$

4.5.3 Functioning test. The booster shall be assembled into a suitable test fixture, either approved Government Furnished Test Equipment or an approved contractor submitted design, and functioned so as to penetrate a one inch aluminum plate. Standoff distance from the aluminum witness plate shall be 9.50 nominal. The aluminum shall be alloy 2014, T4 with alloy 2024, T6 as an alternate.

5. PACKAGING

5.1 Booster auxiliary PA-500. Packaging shall be in accordance with Dwg. 9326028.

5.2 Booster Assembly PA-524. Packaging shall be in accordance with Dwg. 8861645.

5.3 Packing and marking.

5.3.1 Booster auxiliary PA-500. Packing and marking shall be in accordance with Dwg. 9326028, MIL-STD-129, and Code of Federal Regulations - Title 49 - Transportation parts 100-199.

5.3.2 Booster assembly PA-524. Packing and marking shall be in accordance with Dwg. 8861646.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 8-Inch M650, RA Projectile.

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6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection procedures or equipment designs for approval - See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAR-I (D), Dover, New Jersery 07801-5001.

6.4 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.6 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to the Commander, US Army Armament Research and Development Center, Dover, New Jersey 07801-5001, ATTN: AMSMC-QAR-Q (D) for review and approval upon implementation of this Military Specification and subsequently at each 6 month period of production.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-63141B	PROJECTILE 8-INCH, HE, RA, M650

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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